Technical Discussion Q&A Preparation

Suggested structure for the 10-minute follow-on discussion

Answer pattern: state what is known, identify the limit, then name the evidence needed. Do not overstate the concept as validated. A concise engineering answer is usually: calculation → assumption → validation action.

Likely questions and model answers

Why did you choose a 2× static torque factor?

Answer: It is a preliminary target convention for this assignment, not a formal safety factor. The brief omits acceleration, reflected inertia, impacts, stops, duty and life. I would replace the factor-based target with a trajectory- and risk-derived peak/RMS requirement before selection freeze.

Does the motor actually meet the torque and speed simultaneously?

Answer: Not yet proven. The speed ratio implies 4095 RPM, close to the motor's listed 4470 RPM nominal speed. A loaded torque-speed curve, supply/controller voltage limits, temperature, actual motor/gearhead configuration and drive efficiency are required. The sensitivity arithmetic is screening only.

Why does 180 W differ from nominal torque times speed (~198 W)?

Answer: The manufacturer's product label and the listed nominal torque/speed do not multiply to the same value. I would not speculate about rating convention or curve point. I would ask the vendor to confirm the applicable data sheet, continuous operating point and duty conditions.

Is the published 0.321 rpm/mNm speed/torque gradient a contradictory linear slope?

Answer: Not necessarily. Maxon support explains that EC-i iron-winding motors do not have one constant straight-line speed/torque characteristic; the gradient depends on speed. I would request/use the actual motor curve and not extrapolate that single gradient as a constant slope. The separate 180 W label versus nominal Tω arithmetic still needs the applicable rating point clarified.

Why use a gearhead with 1° backlash?

Answer: It is a candidate for torque and ratio screening, not an unconditional choice for accurate humanoid motion. At 350 mm, 1° maps to about 6.1 mm arc. If the position-error budget is tighter, use a lower-backlash/stiffer reducer or output feedback, then include compliance and control performance in the full error budget.

Why not let the gearbox output support the forearm?

Answer: The gearbox torque rating is not a cantilever radial-load rating. The concept uses an independent spindle supported by two spaced bearings and routes those reactions into the split-yoke and upper arm. Shaft, bearing, fit and carrier calculations are still required.

What changed in your structural screen after the CAD review?

Answer: I used the current geometry coordinates: bearing centres at Y=11 and 61 mm and forearm centre plane Y=111 mm. That makes 50 mm spacing and 50 mm overhang. The 1.22625 N·m moment is converted explicitly to 1226.25 N·mm before the mm-based stress formula; the resulting nominal bending stress is 7.23 MPa.

Are your shaft stress numbers a proof of safety?

Answer: No. They are nominal static smooth-shaft screens. They exclude keyway/shoulder stress concentrations, cyclic fatigue, impact, manufacturing condition, deflection, actual hub contact and uncertain link loads. A detailed drawing and worst-case dynamic loads are needed.

Why use two 6201 bearings?

Answer: They provide a separated support load path in a compact concept. The envelope dimensions align to the nominal 12 mm spindle and 32 mm housing seat. This does not demonstrate fit, oscillatory bearing life, preload, axial loading, lubrication or housing stiffness; those are verification gates.

What are the two biggest requirements you would ask the project lead?

Answer: First: full duty/motion profile including acceleration, reversals, impacts, dwell and life because that determines peak/RMS torque and thermal loading. Second: output accuracy/backlash/stiffness and power-loss hold/safe-state requirements because they can change reducer, feedback, brake and controls architecture.

Why no brake?

Answer: The brief does not define an unattended gravity-hold requirement. This baseline is a supervised concept and must be externally supported when de-energized. If it must hold the arm after power loss, a normally engaged brake or mechanical counterbalance/support becomes a release requirement and must be sized and safety-reviewed.

What would you test first after a prototype is authorized?

Answer: First freeze and verify the configured components and safe-state architecture. Then use a guarded, restrained bench, current limits, E-stop, calibrated torque and RPM measurement, and temperature logging. Start with unloaded and low-load checks, ramp to 45 RPM at defined torque, then run the representative duty cycle and measure backlash/deflection. Predefine abort thresholds.

Did you run FEA or a real prototype test?

Answer: No. The brief specifically says prototype and detailed FEA are not required, and no hardware was purchased or made. The package contains a CAD envelope model and a simulation/verification plan; it does not include invented solver output or bench data.

Where is the native FreeCAD file?

Answer: FreeCAD was not installed in the generation environment. The deliverable includes a neutral STEP model, CadQuery builder source, and a FreeCAD macro. Open/run the macro inside FreeCAD and save .FCStd locally; the STEP file is the CAD-equivalent exchange deliverable allowed in the brief.

What does the CAD validation tell you?

Answer: The STEP re-imported through CadQuery and reported valid geometric solids with 21 solids in the assembly. This checks file geometry validity only. It does not prove no interference, kinematics, tolerance fit, stress, bearing behavior, or manufacturability.

What if the actual efficiency is 30%?

Answer: The simplified motor-side requirement rises to about 0.446 N·m and 191 W, which exceeds the listed nominal torque/current screen and 180 W class label. That is a no-go for the simplified continuous requirement unless the actual loaded curve and duty demonstrate acceptable operation; I would investigate another motor/ratio/gearhead and confirm with the supplier.

Suggested 10-minute discussion pacing

Time	Focus	Goal
0:00–2:00	Torque calculation / margin	Show calculation discipline and assumption boundary.
2:00–4:00	Drive selection / motor power caveat	Show manufacturer evidence and no bluffing about open point.
4:00–6:00	Load path / shaft support / backlash	Explain why torque and radial support are different.
6:00–8:00	Risk, safe state, dynamic/thermal validation	Show awareness of missing system requirements.
8:00–10:00	CAD/file status and next steps	State what geometry check proves, and what remains to close.